DIGITAL EGYPT PIONEERS INITIATIVE (DEPI)

Multi-Source Job Market Data Analysis

Transforming raw job postings into evidence-based career intelligence. Team: NHA-5-S2-104

Problem Statement

The core market challenge for modern job seekers and training providers

Scattered Platforms

Job listings fragmented across multiple platforms globally.

Inconsistent Data

Mismatched job titles, skill names, and salary formats.

Unclear Demand

Advisors struggle to identify truly high-demand technical skills.

Scope & Data Sources

Aggregating robust datasets for comprehensive market intelligence

11,000+

Unified job postings processed through 18 structured columns.

Adzuna API

General job search, salaries, and labor-market statistics.

Remote OK & Board APIs

Remote-only feeds and company-direct Greenhouse & Lever endpoints.

Architecture & Tech Stack

End-to-end data pipeline from extraction to executive dashboards

Medallion Architecture

SQL Server Bronze (Raw) to Silver (Cleaned) and Gold (Star Schema).

Python Processing

pandas, NumPy, SQLAlchemy, and Seaborn for automated EDA and cleaning.

BI & Reporting

Power BI & Tableau interactive dashboards with robust DAX measures.

Core Business Questions

Key analytical dimensions driving our data modeling

Role Demand & Volume

Tracking highest-demand roles and posting volume trends over time.

Skill Clustering

Identifying technical and soft skills that frequently co-occur by role.

Salary Benchmarks

Analyzing advertised annual/hourly compensation by geography and sector.

Remote Work Dynamics

Evaluating remote-first vs. direct company postings and benefits.

PROJECT LEADERSHIP

Team Members & Roles

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Project Manager & QA Lead

Shehab Talaat Ismail
Data Engineer Lead (W1)

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